Internal Stage

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Snowflake Snowpro Certified

Section Introduction-Internal Stages

- What is an Internal Stage?
- What are the types of Internal Stages supported in Snowflake ?
- What is SnowSQL ?
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- LAB-2-Login to Snowflake using SnowSQL.
- LAB-3-Named Internal Stage-PUT.
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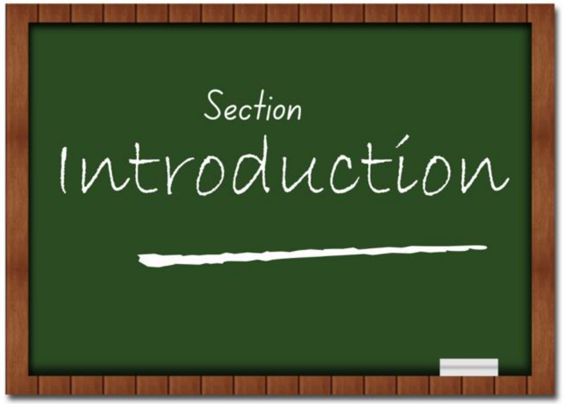
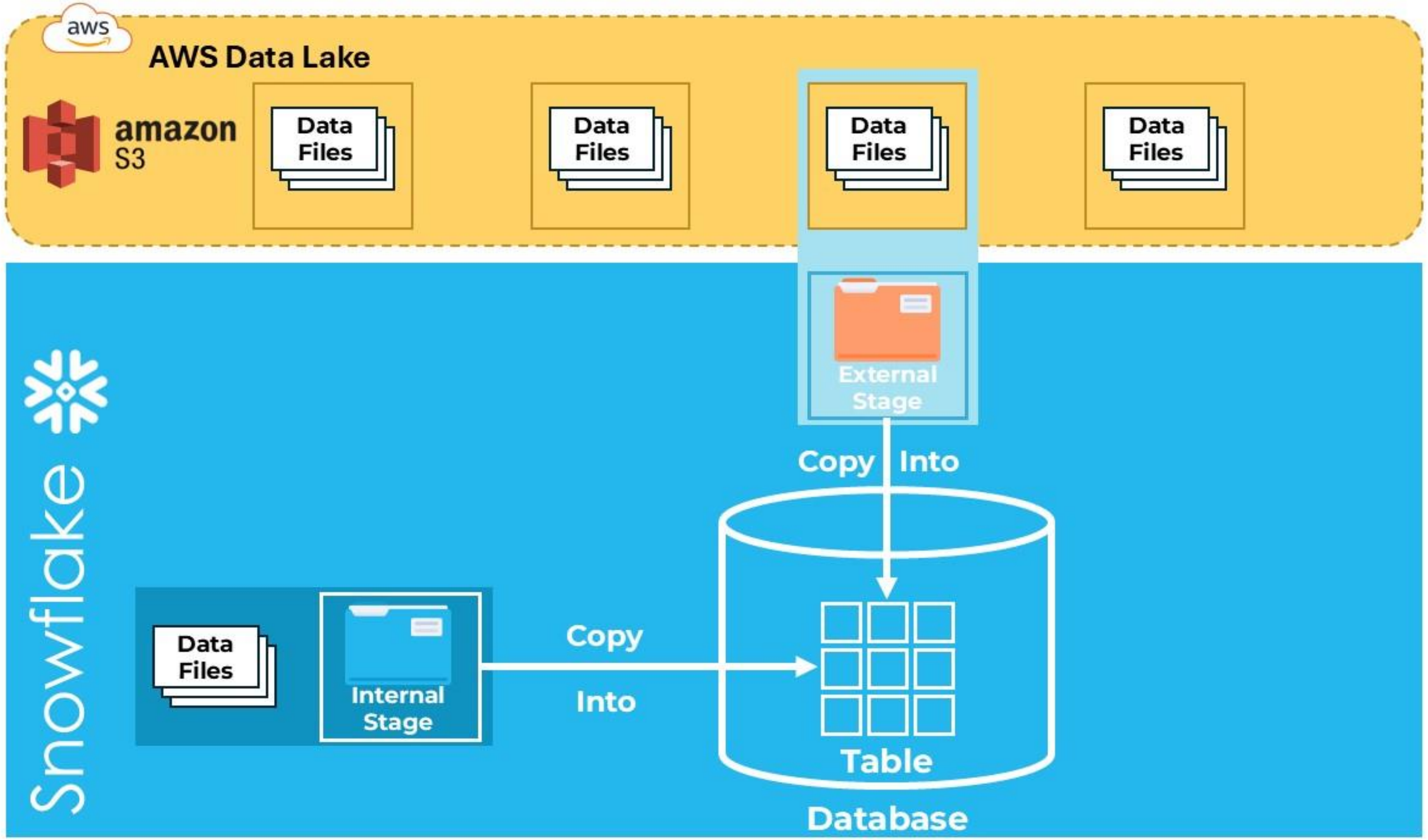
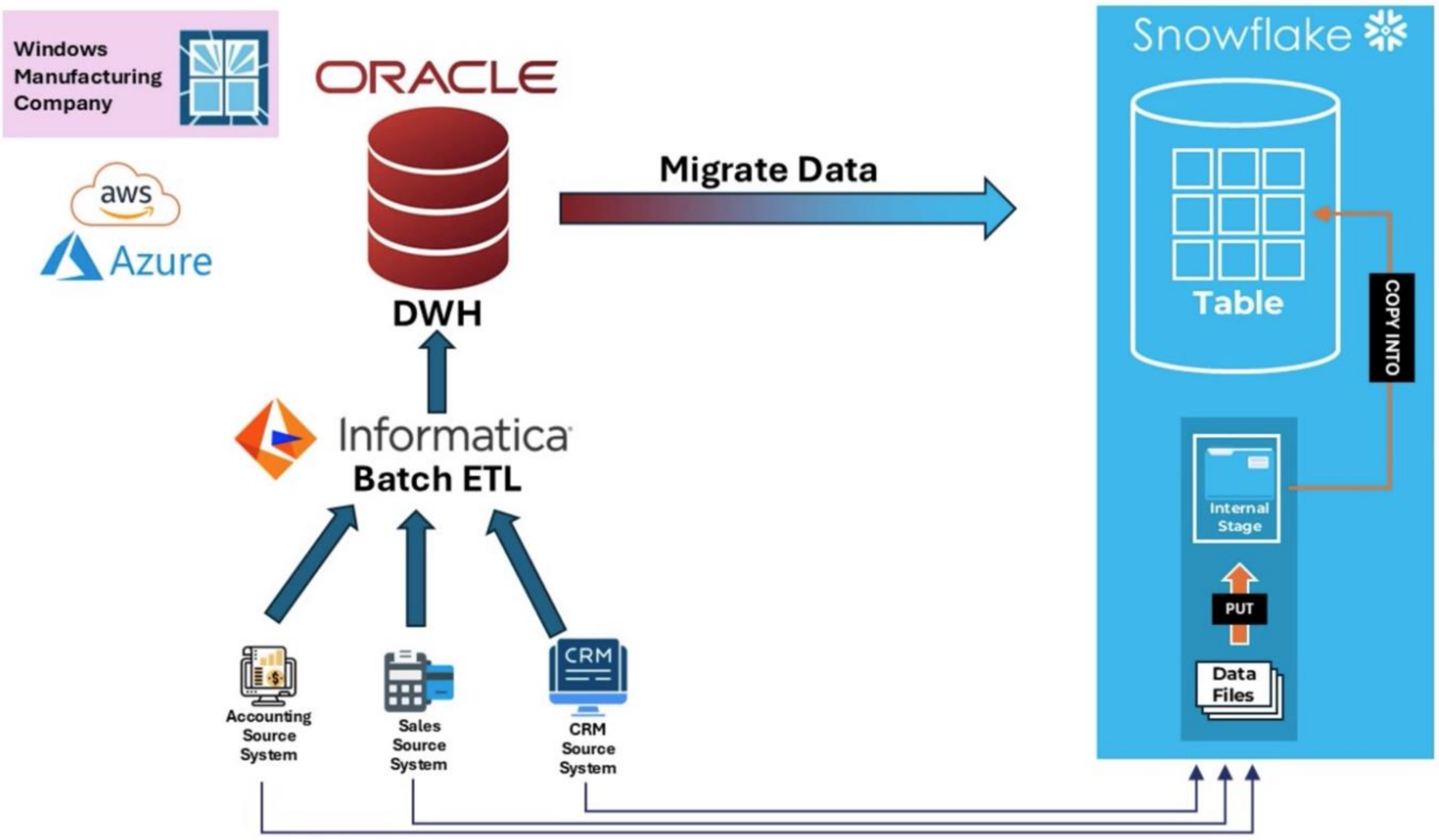
- LAB-5-Named Internal Stage-REMOVE.
- LAB-6-Working with USER STAGE.
- LAB-7-Working with TABLE STAGE .
- LAB-8-Loading Data to & Unloading Data from Internal Stage.
- Summary.



What is an INTERNAL STAGE?

An Internal stage in Snowflake is a secure storage location within your Snowflake account where you can store data files for loading data to Snowflake.

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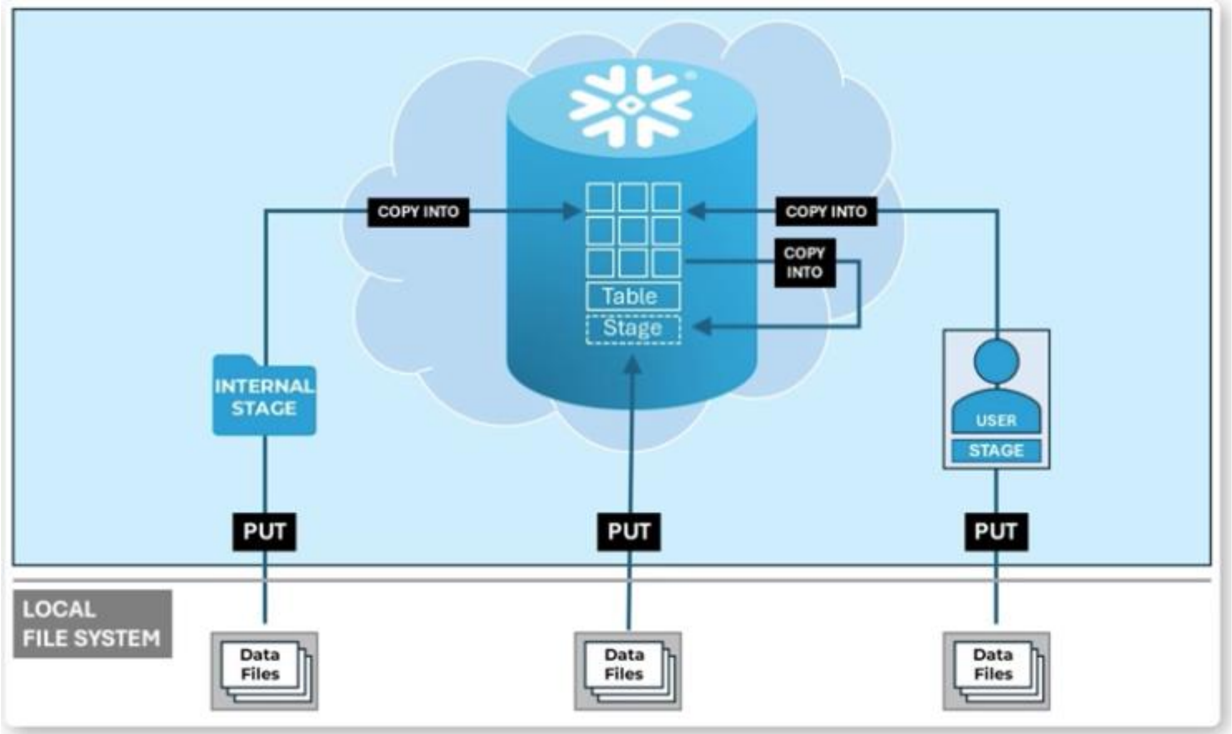
System
CRM
CRM
Source
System
Snowflake
Table
COPY INTO
Internal
Stage
PUT
Data
Files

What are the types of Internal Stages supported in Snowflake

There are 3 types of Internal Stages supported in Snowflake.

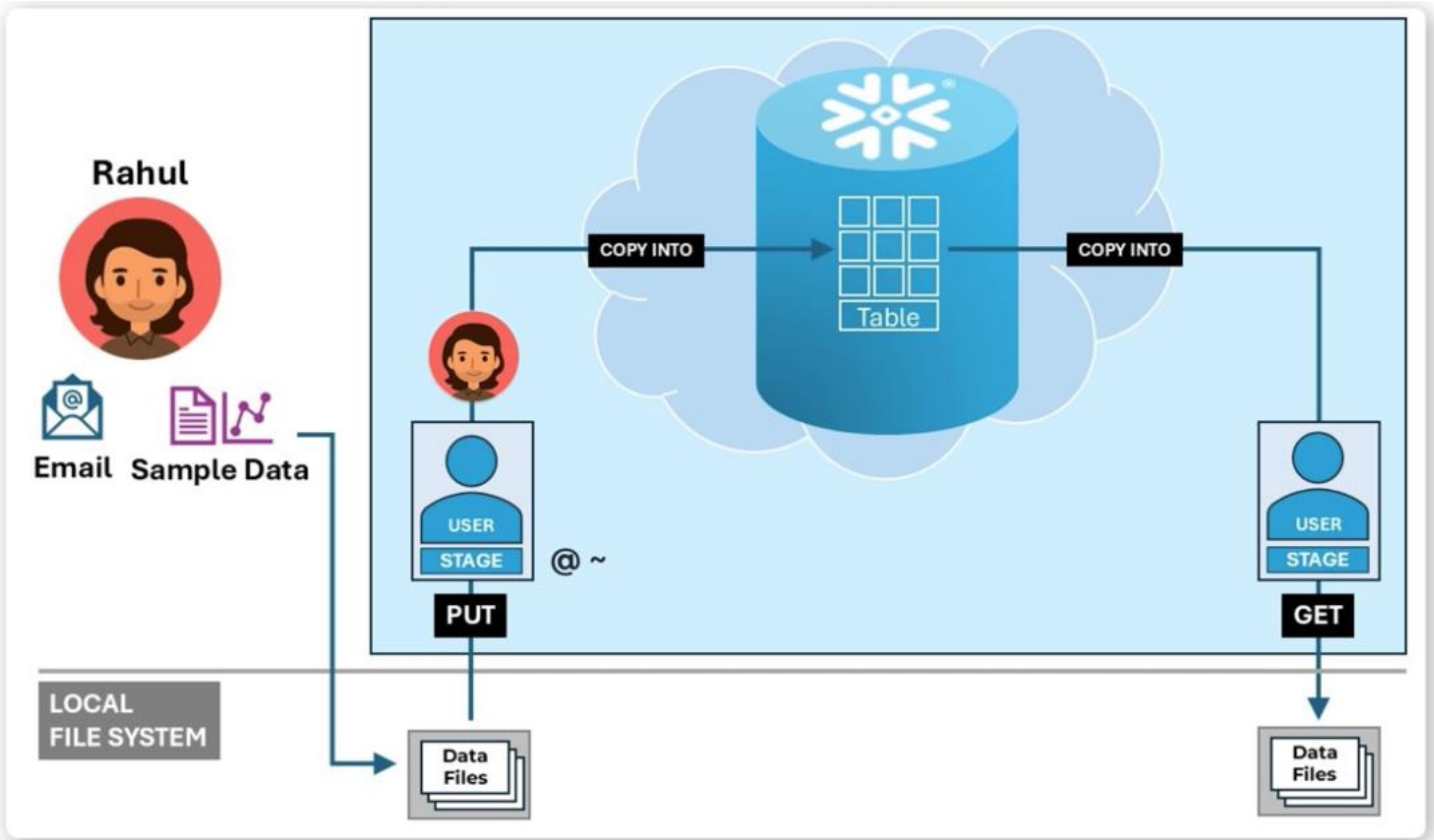
- User Stage
- Table Stage
- Named Internal Stage



What are the types of Internal Stages supported in Snowflake

User Stage

The User Stage is a secure storage location within your Snowflake account which is automatically created when a user is created in Snowflake.



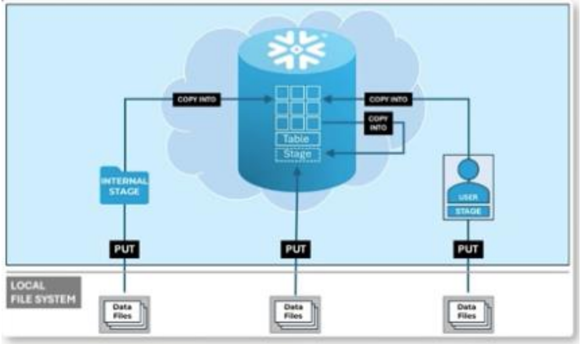
What are the types of Internal Stages supported in Snowflake

User Stage

The User Stage is a secure storage location within your Snowflake account which is automatically created when a user is created in Snowflake.

Features of User Stage

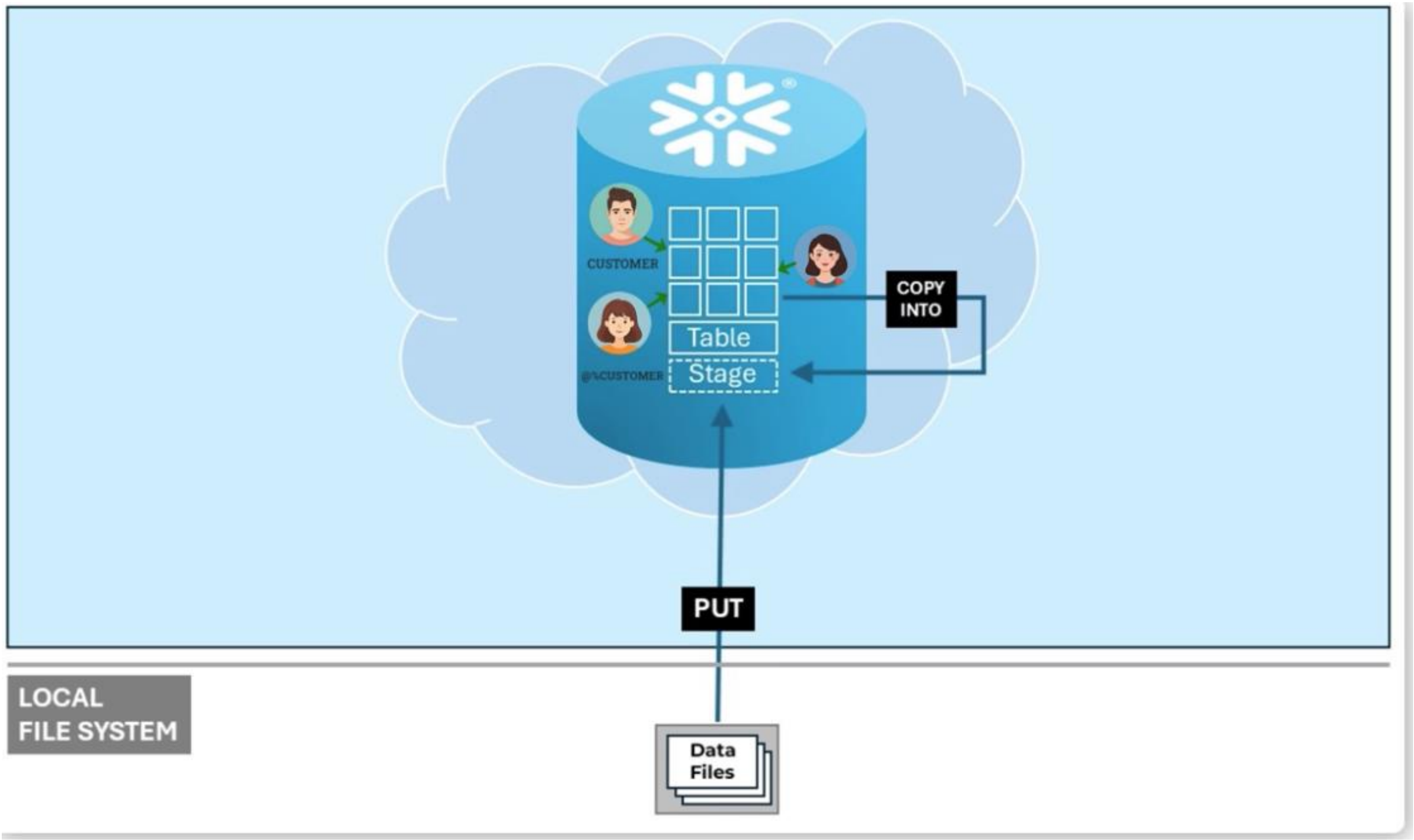
- Automatically created when a user is created in Snowflake.
- User Stage cannot be altered or dropped without dropping the user.
- User has complete privileges on the stage which cannot be taken away and Is accessible to the user without the need to grant privileges on the stage.
- Cannot be used by any other user or role.
- Private sandbox area for the user where they can upload files of their choice.
- User Stage is represented by @~ .
- Files in the stage can be listed with LIST @~.
- User Stage can be used with the COPY statement to load tables in Snowflake.



What are the types of Internal Stages supported in Snowflake

Table Stage

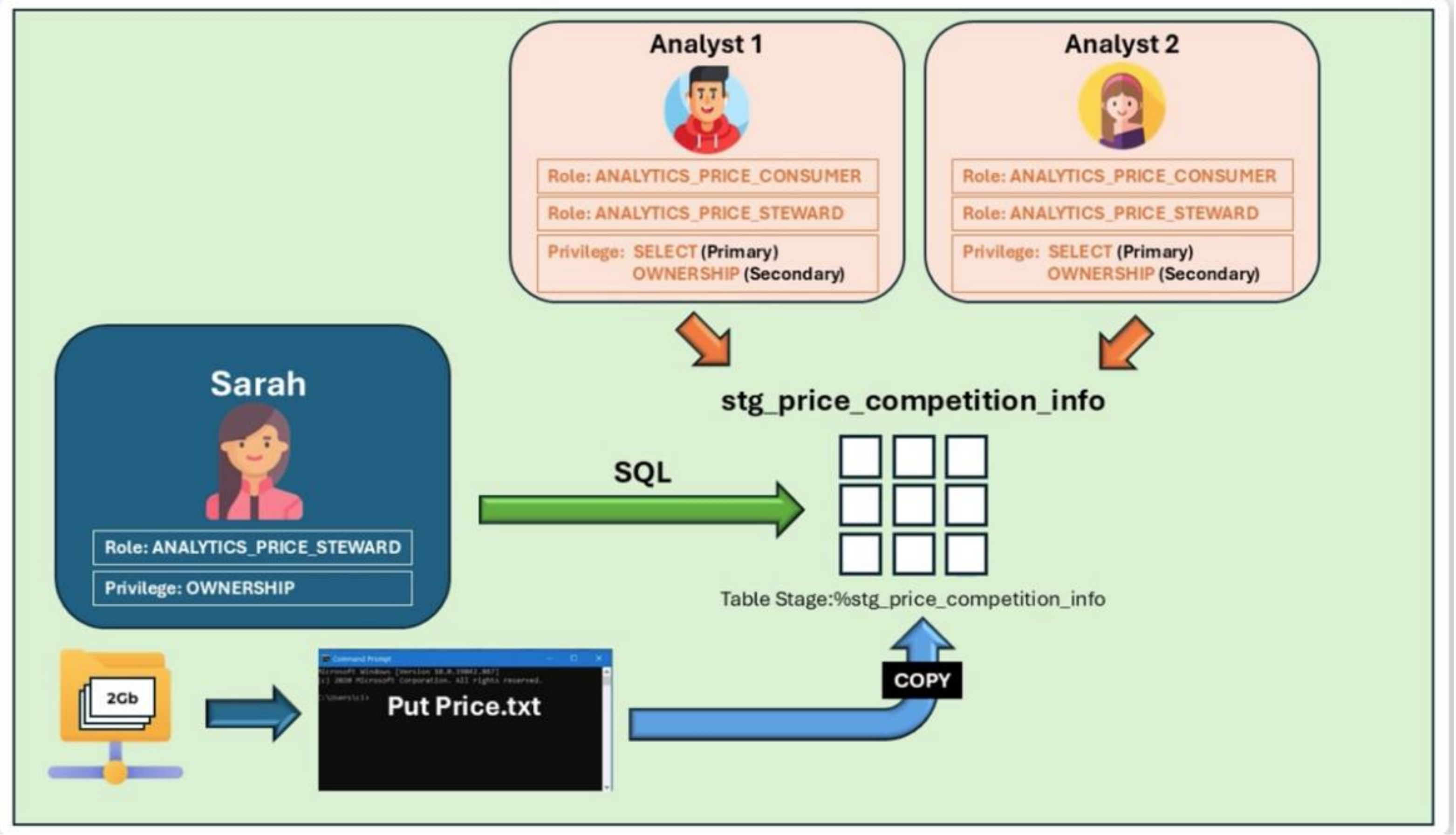
The Table stage is a secure storage location within your Snowflake account which is automatically created when a table is created in Snowflake.



What are the types of Internal Stages supported in Snowflake

Why use Table Stage ?

- Automatic & Table Specific.
- Ownership Based File management.
- Simplicity for the owner.



What are the types of Internal Stages supported in Snowflake

Table Stage

The Table stage is a secure storage location within your Snowflake account which is automatically created when a table is created in Snowflake.

Features of Table Stage

- Is automatically created when a table is created in Snowflake.
- Cannot be dropped independently from the table.
- Is dropped when the table is dropped.
- Table owner has complete privileges on the stage which cannot be taken away.
- Table Stage is referenced by @%Tablename.
- Files in the stage can be listed with LIST @%Tablename.
- Table stage can be used with the COPY statement to load data to its linked tables in Snowflake, it cannot load to any other table.

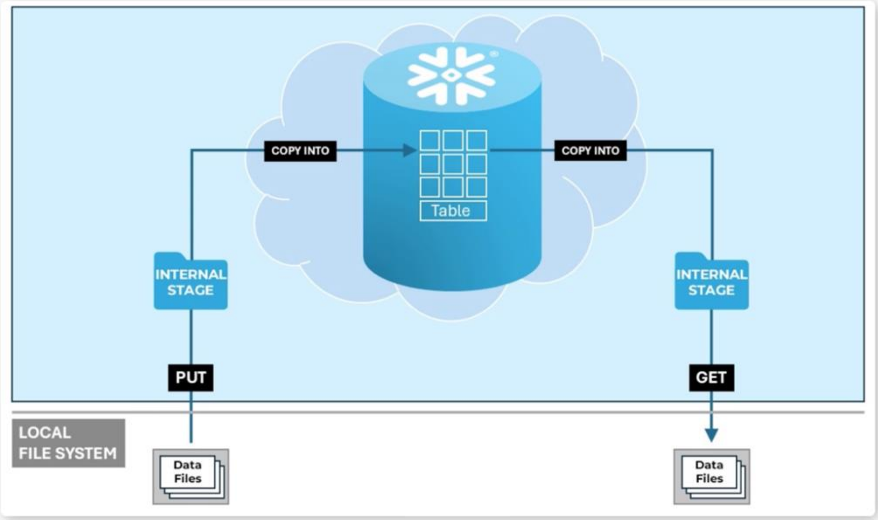
What are the types of Internal Stages supported in Snowflake

Named Internal Stage

The Named Internal stage is a secure storage location within your Snowflake account which must be explicitly created with the CREATE OR REPLACE STAGE statement.

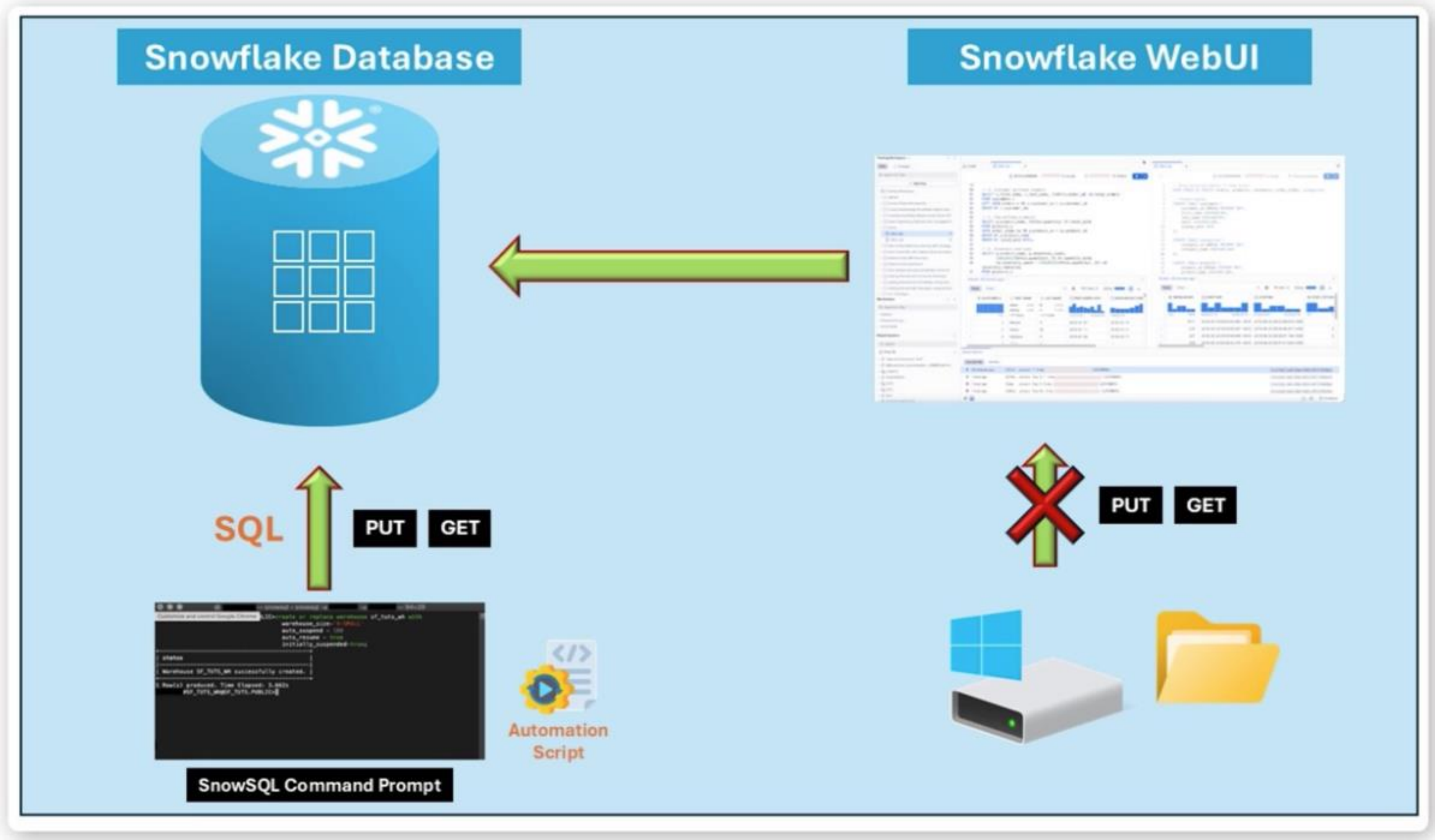
Features of Named Internal Stage

- Must be explicitly created with CREATE OR REPLACE STAGE statement.
- It is not hindered by the limitations of user or table stages which ties it to a specific User or table.
- Access to a stage must be granted explicitly to be able to use the stage.
- The files in the stage can be accessed with LIST @<stagename>
- Stages do not need storage integration or URL information.



What is SnowSQL ?

SnowSQL is a Command-Line Interface (CLI) tool/client software for connecting to Snowflake.
It is a terminal-based tool to interact with your Snowflake database without web-based Snowflake UI



What is SnowSQL ?

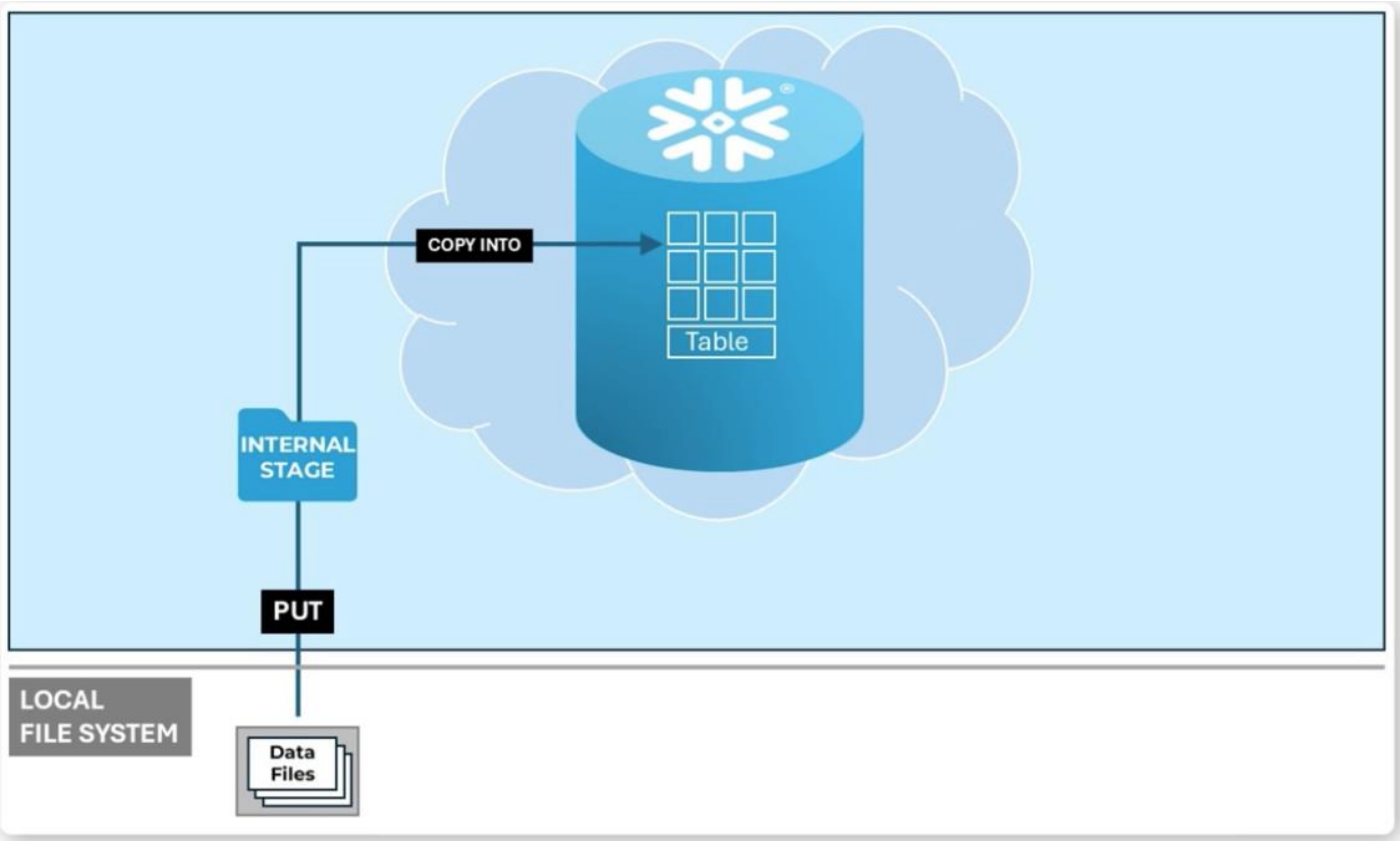
The CLI commands supported by SnowSQL

- PUT
- GET
- REMOVE

SQL Commands supported to interact with Internal Stages

PUT

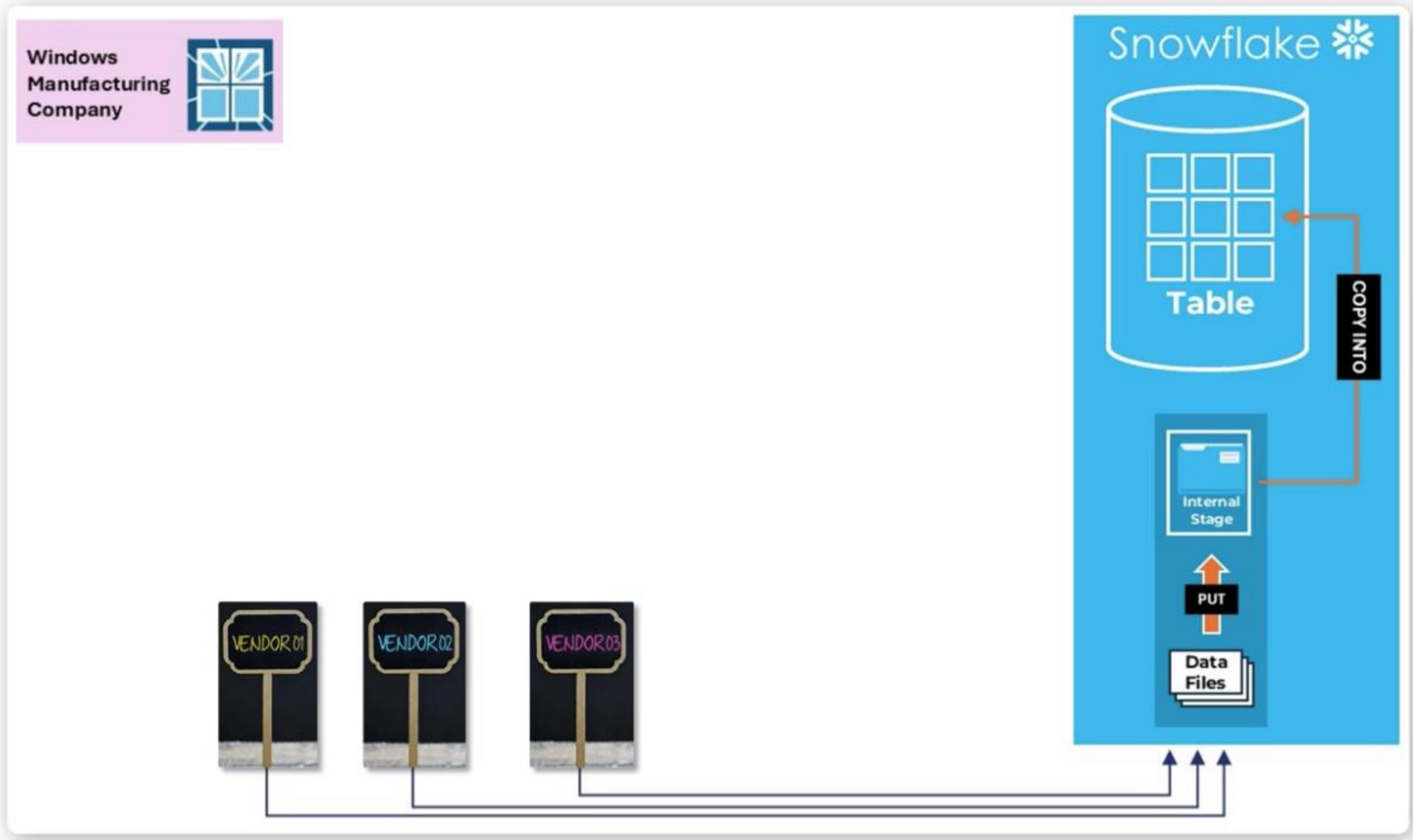
PUT Uploads one or more data files from a local file system onto an internal stage.



SQL Commands supported to interact with Internal Stages

PUT

PUT Uploads one or more data files from a local file system onto an internal stage.



SQL Commands supported to interact with Internal Stages

PUT

PUT Uploads one or more data files from a local file system onto an internal stage.

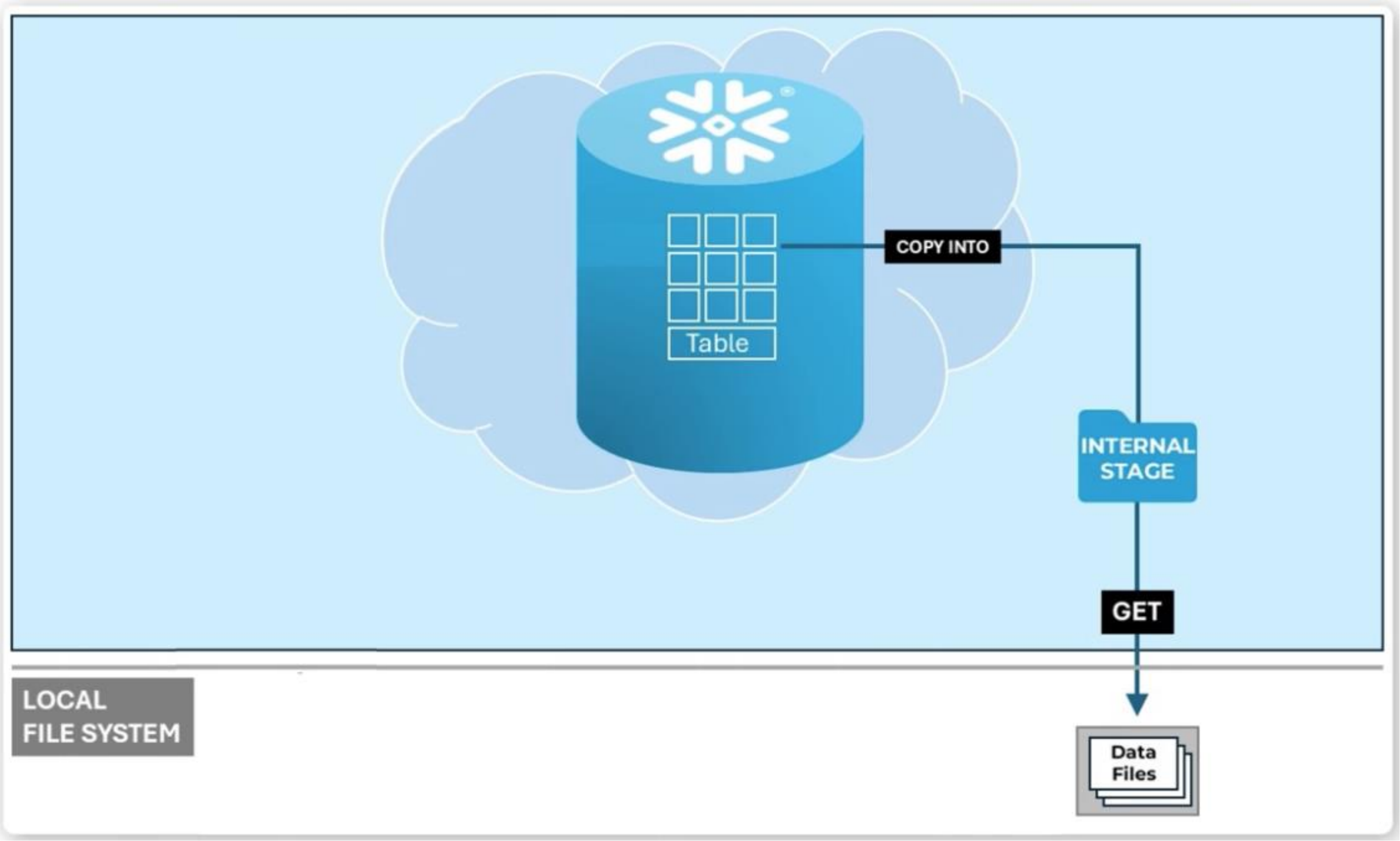
Features of PUT

- PUT can upload files to all 3 types of Internal stages.
- PUT does not support uploading files to external stages.
- PUT skips any file that is already present at the target location.
- PUT uploads files in parallel by default.
- PUT compresses the file before it is uploaded to the stage.
- PUT encrypts files during upload.
- OVERWRITE=TRUE parameter can be used to replace existing files in the stage.

SQL Commands supported to interact with Internal Stages

GET

GET downloads one or more data files to your local file system from an internal stage.



SQL Commands supported to interact with Internal Stages

GET

GET downloads one or more data files to your local file system from an internal stage.

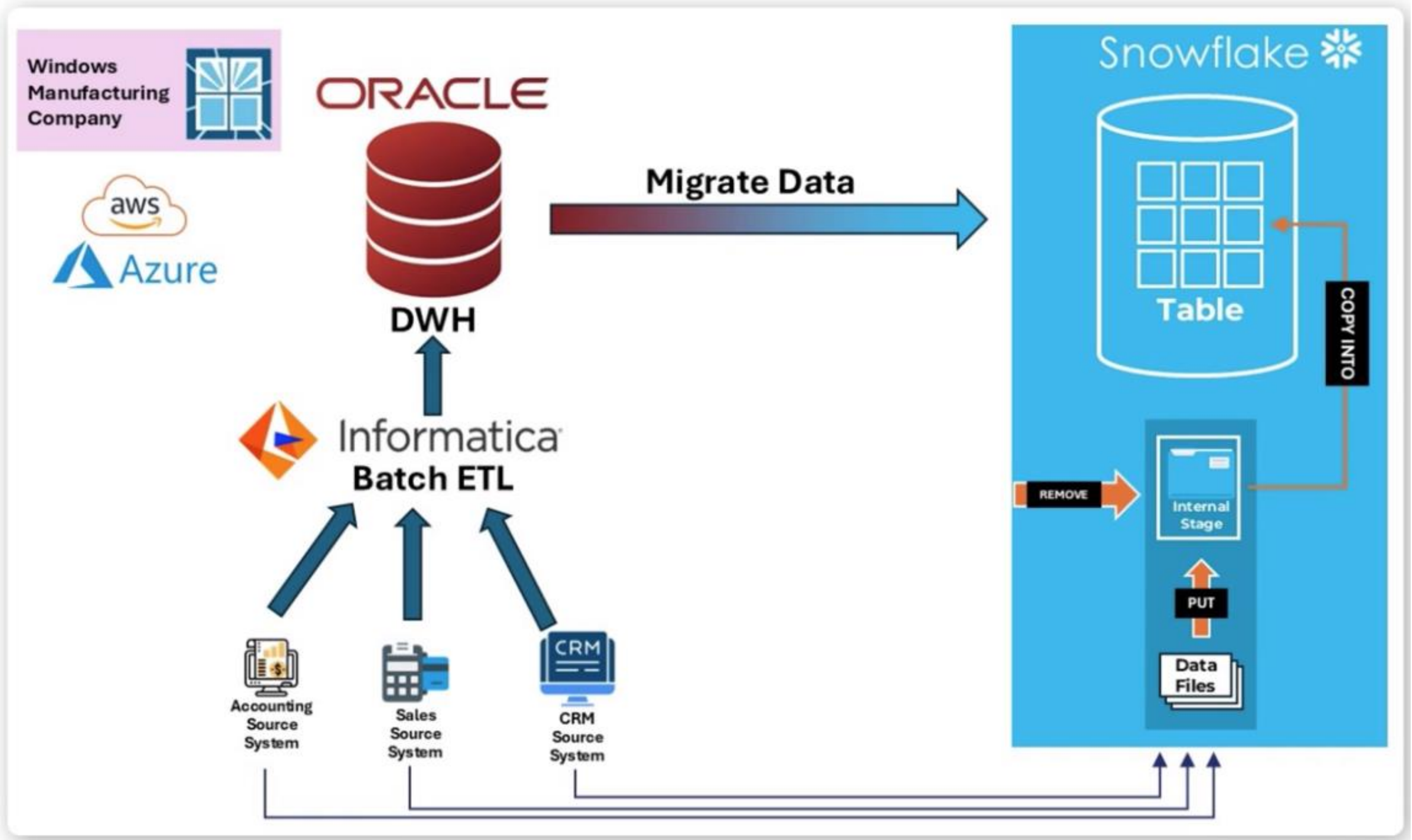
Features of GET

- GET can download files from all 3 types of Internal stages.
- GET does not support downloading files from external stages.
- GET does not compress or uncompress the file before it is downloaded.
- Files are downloaded in parallel by default with GET.

SQL Commands supported to interact with Internal Stages

REMOVE

REMOVE removes one or more data files from the stage.



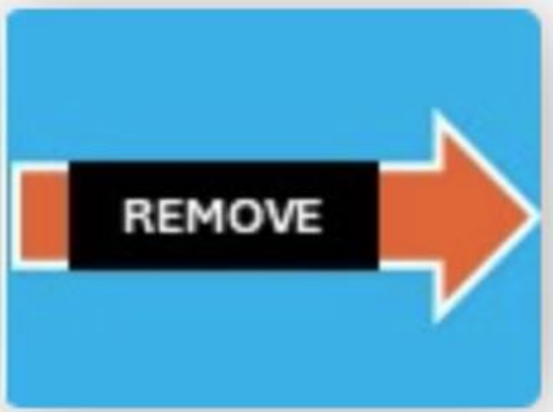
SQL Commands supported to interact with Internal Stages

REMOVE

REMOVE removes one or more data files from the stage.

Features of REMOVE

- REMOVE can remove files to all 3 types of Internal Stages
- REMOVE can remove files from external stages.



Section Summary

- An Internal stage in Snowflake is a secure storage location within your Snowflake account where you can store data files for loading data to Snowflake.
- There are 3 types of Internal stages supported in Snowflake.
  - User Stage
  - Table Stage
  - Named Internal Stage
- A User Stage is a secure storage location within your Snowflake account which is automatically created when a User is created in Snowflake.
- A User Stage cannot be altered or dropped without dropping the user.
- User has complete privileges on the stage which cannot be taken away and is accessible to the user without the need to grant privileges on the stage.
- User Stage cannot be used by any other user or role.
- User Stage operates like a private sandbox area where the user can upload files of their choice.
- User Stage is represented by @~.
- Files in the User Stage can be listed with LIST @~.
- User Stage can be used with the COPY statement to load to and unload from tables in Snowflake.
- Table stage is a secure storage location within your Snowflake account which is automatically created when a table is created in Snowflake.
- Table stage cannot be dropped independently from the table and is dropped when the table is dropped.

Section Summary

- Table owner has complete privileges on the stage which cannot be taken away.
- Table Stage is referenced by @%Tablename.
- Files in the Table stage can be listed with LIST @%Tablename.
- Table stage can be used with the COPY statement to load data to or unload data from its linked tables in Snowflake, it cannot load to or unload from any other unrelated table.
- The Named Internal Stage is a secure storage location within your Snowflake account which must be created with the CREATE or REPLACE STAGE statement.
- It is not constrained by the limitations of USER /TABLE STAGE which ties it to a specific user or table.
- Access to the Named Internal Stage must be granted explicitly to be able to use the stage.
- Files in the Named Internal stage can be accessed with LIST @<stagename>.
- Named Internal Stages do not need Storage integration or URL information.
- SnowSQL is a command-Line Interface(CLI) tool/client software for connecting to Snowflake.
- SnowSQL is a terminal-based tool to interact with your Snowflake database without web-based Snowflake UI.
- SnowSQL support 3 different types of commands to interact with Internal Stages.
    - PUT
    - GET
    - REMOVE

Section Summary

- PUT uploads one or more data files from a local file system onto an Internal Stage.
- PUT can upload files to all 3 types of Internal Stages.
- PUT does not support uploading files to an External Stage.
- PUT encrypts file during upload.
- PUT skips any file that is already present at the target location.
- PUT uploads files in parallel by default. You can provide an integer value to set the parallelism.
- PUT compresses the file by default before it is uploaded to the stage.
- OVERWRITE=TRUE parameter can be used to replace existing files in the stage.
- SOURCE_COMPRESSION must be provided if your file is compressed. The Default is AUTO_DETECT.
- GET downloads one or more data files to your local file system from an Internal Stage.
- GET can download files from all 3 types of Internal Stages.
- GET does not support downloading files from External Stages.
- GET does not compress/uncompress the file before it is downloaded, it downloads the file as-is.
- GET downloads files in parallel by default. You can provide an integer value to set the parallelism.
- PATTERN can be provided to selectively download files from the stage.
- REMOVE removes one or more data files from the stage.
- REMOVE can remove files from all 3 types of Internal Stages.
- REMOVE can remove files from External Stages.
- REMOVE can be abbreviated as RM.
- PATTERN can be provided to selectively remove files from the stage.

Section Summary

- When connecting to Snowflake via SnowSQL use Account Locator instead of Account Name.
- Account Locator format will be different depending on the Snowflake region.
- All SQL statements that work on the Snowflake Web-UI work via SnowSQL.
- PUT,GET,REMOVE cannot be used on the Snowflake Web-UI.
- Internal Stages can be partitioned during upload by adding a directory to the upload path in PUT.
- Use COPY to Load and Unload data to and from Internal stages.
- All COPY options that work with External Stages work with COPY on INTERNAL Stages.

Thank you

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